

**2.43 describe experiments to investigate the effect of light on net gas exchange from a leaf, using hydrogen-carbonate indicator**

**Humans**

- 2.44 describe the structure of the thorax, including the ribs, intercostal muscles, diaphragm, trachea, bronchi, bronchioles, alveoli and pleural membranes
- 2.45 understand the role of the intercostal muscles and the diaphragm in ventilation
- 2.46 explain how alveoli are adapted for gas exchange by diffusion between air in the lungs and blood in capillaries
- 2.47 understand the biological consequences of smoking in relation to the lungs and the circulatory system, including coronary heart disease
- 2.48 describe experiments to investigate the effect of exercise on breathing in humans.

## **h) Transport**

*Students will be assessed on their ability to:*

- 2.49 understand why simple, unicellular organisms can rely on diffusion for movement of substances in and out of the cell
- 2.50 understand the need for a transport system in multicellular organisms

### **Flowering plants**